



**Year 11 Mathematics Advanced
Assessment Task 2
Term 3 Week 3, 2023**

Setter: EKA

Name: _____

Teacher: _____

- Answer Questions in the space provided.
- Answer multiple choice questions on the question paper by circling the correct alternative.
- Show all relevant mathematical reasoning and/or calculations.
- Use blue or black PEN only.
- NESA approved calculators are permitted in this task.
- Marks may be deducted for insufficient or illegible work.
- Extra writing space is provided at the back of this booklet.
If you use this space, clearly indicate which question you are answering.
- Total possible marks: **32**
- Time allowed: **45 minutes plus 2 minutes Reading Time**

Wednesday, 2nd August 2023

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
/6	/6	/4	/4	/4	/4	/4	/32

Question 1 (6 marks)

Marks

- (a) Circle your correct alternative.

1

The graph of the function $f(x) = 18$ intersects with the graph of $g(x) = 2^x$ where:

(A) $x = \log_2 18$

(B) $x = \frac{1}{18}$

(C) $x = 9$

(D) $x = \log_{18} 2$

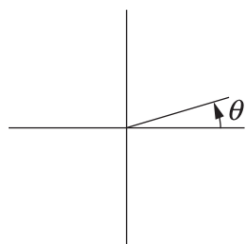
- (b) Circle your correct alternative.

1

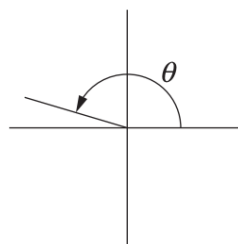
For the angle θ , $\sin \theta = \frac{7}{25}$ and $\cos \theta = -\frac{24}{25}$.

Which diagram best shows the angle θ ?

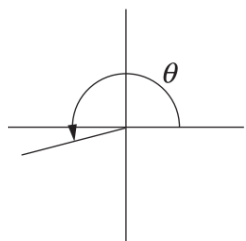
(A)



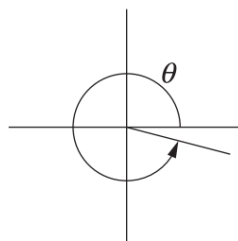
(B)



(C)



(D)



- (c) Solve $\cos \theta = 0.63$ for $-180^\circ \leq \theta \leq 0^\circ$, giving your answer to the nearest degree.

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Question 1 Continued

- (d) Find the derivative of $y = \frac{3x^4 - 1}{x}$.

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- (e) Given that $f(x) = x^2 + 1$ and $g(x) = \sqrt{x - 3}$. Find $y = f(g(x))$ and state its domain.

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End of Question 1

Question 2 (6 marks)

Marks

- (a) Circle your correct alternative.

1

What is the exact value of $\cos 135^\circ + \operatorname{cosec} 60^\circ$?

(A) $\frac{2\sqrt{2} - \sqrt{3}}{\sqrt{6}}$

(B) $\frac{2\sqrt{2} - 1}{\sqrt{2}}$

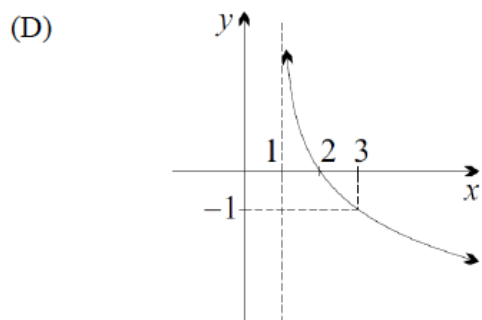
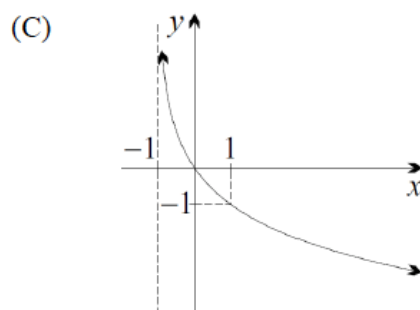
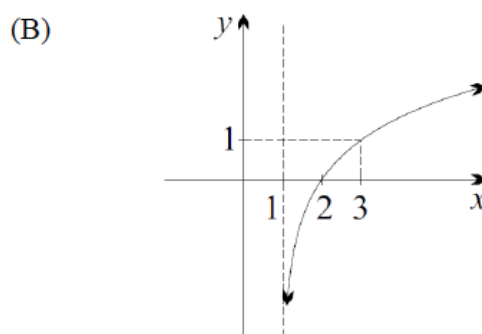
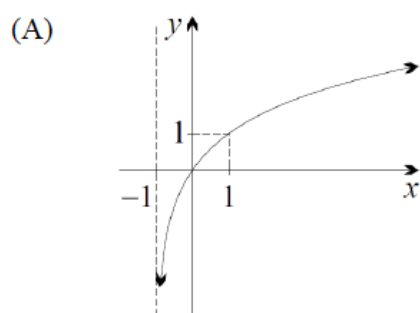
(C) $\frac{2\sqrt{2} + \sqrt{3}}{\sqrt{6}}$

(D) $\frac{2\sqrt{2} + 1}{\sqrt{2}}$

- (b) Circle your correct alternative.

1

Which of the following is the graph of $y = -\log_2(x+1)$?



Question 2 Continued

- (c) Solve the equation $|2x - 3| = 6$.

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- (d) Differentiate with respect to x . Leave your answer in fully factorised form.

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$$y = \frac{2x^2}{2 - x}$$

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End of Question 2

Question 3 (4 marks)**Marks**

(a) Solve $\sin\left(\frac{x}{2}\right) = \frac{1}{2}$ for $0^\circ \leq x \leq 360^\circ$.

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(b) Show that $\cot A + \frac{\sin A}{1 + \cos A} = \operatorname{cosec} A$.

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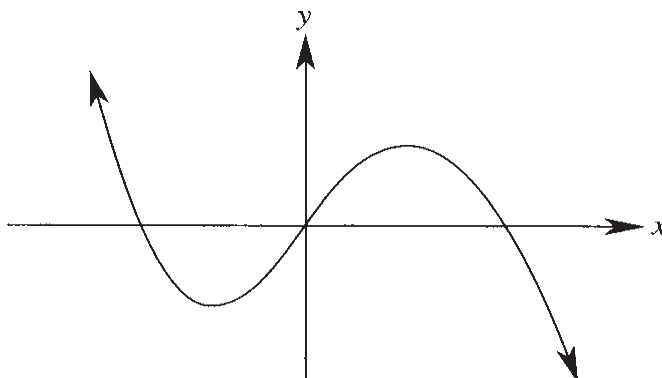
End of Question 3

- (a)
- i) Show that the derivative of $f(x) = x^2 + 3x + 5$ is $f'(x) = 2x + 3$ by **first principles**.
- 2

- ii) Hence, find the equation of the normal to the curve $f(x) = x^2 + 3x + 5$ at the point $(1, 9)$.
- 2

Question 5 (4 marks)**Marks**

- (a) The diagram below shows a sketch of the function $y = f(x)$.
On the same diagram sketch the gradient function $y = f'(x)$.

1

- (b) Show that, $\frac{d}{dx} [3x^2(4x - 1)^3] = 6x(10x - 1)(4x - 1)^2$.

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End of Question 5

Question 6 (4 marks)**Marks**

- (a) Differentiate with respect to x .

$$y = \sqrt{x^2 - 2x}$$

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- (b) Solve $(\log_2 x)^2 - 8 = \log_2 x^2$.

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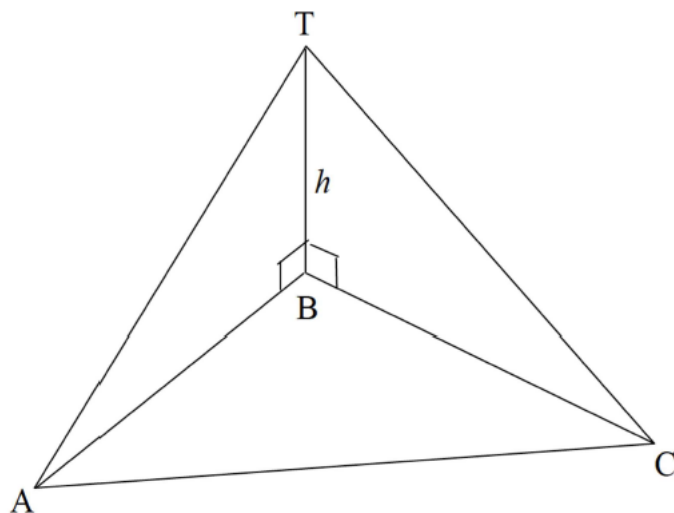
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End of Question 6

The angle of elevation of a vertical tower BT of height h metres, at a point A due west of it, is 14° . From the point C , the bearing of the tower is 343° and the angle of elevation is 11° . The points A and C are 1000m apart and on the same level as the base B of the tower.



- (a) By drawing a diagram showing the bearings, explain why $\angle ABC = 107^\circ$.

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- (b) Show that $AB = h \tan 76^\circ$ and find a similar expression for CB .

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(c) Hence, find the height of the tower, correct to the nearest metre.

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End of Question 7

End of Assessment Task

Extra writing space

If you are using this space indicate clearly which question(s) you are answering & write ‘see extra writing space’ on the original question space provided for that question.

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines, providing a guide for letter height and placement. The lines are evenly spaced across the entire page, leaving ample room for writing practice. There is no text or other markings on the page.



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Marks

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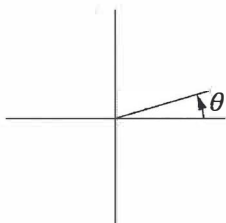
$2^x = 18$
 $x = \log_2 18$

- (b) Circle your correct alternative.

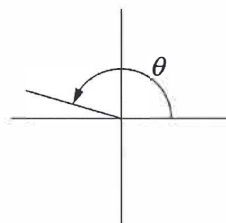
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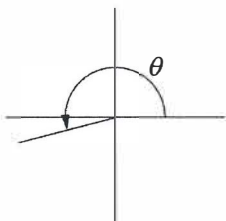
(A)



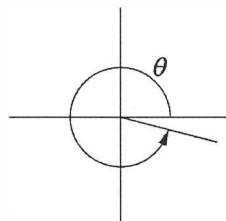
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(C)



(D)



- (c) Solve $\cos \theta = 0.63$ for $-180^\circ \leq \theta \leq 0^\circ$, giving your answer to the nearest degree.

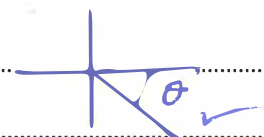
1

$\cos \theta = 0.63$

ref $\angle = \cos^{-1}(0.63)$

$= 51^\circ$

$\theta = -51^\circ$ ✓



Question 1 Continued

- (d) Find the derivative of $y = \frac{3x^4 - 1}{x}$.

1

$$\begin{aligned} y &= 3x^3 - x^{-1} \\ \frac{dy}{dx} &= 9x^2 - (-1 \times x^{-2}) \\ &= 9x^2 + \frac{1}{x^2} \end{aligned}$$

- (e) Given that $f(x) = x^2 + 1$ and $g(x) = \sqrt{x-3}$. Find $y = f(g(x))$ and state its domain.

2

$$\begin{aligned} f(g(x)) &= (\sqrt{x-3})^2 + 1 \\ &= x-3 + 1 \\ &= x-2 \end{aligned}$$

$$\text{domain of } g(x) = x \geq 3$$

$$\therefore \text{domain of } f(g(x)) = x \geq 3$$

End of Question 1

Question 2 (6 marks)

Marks

- (a) Circle your correct alternative.

1

What is the exact value of $\cos 135^\circ + \operatorname{cosec} 60^\circ$?

(A) $\frac{2\sqrt{2} - \sqrt{3}}{\sqrt{6}}$

(B) $\frac{2\sqrt{2} - 1}{\sqrt{2}}$

(C) $\frac{2\sqrt{2} + \sqrt{3}}{\sqrt{6}}$

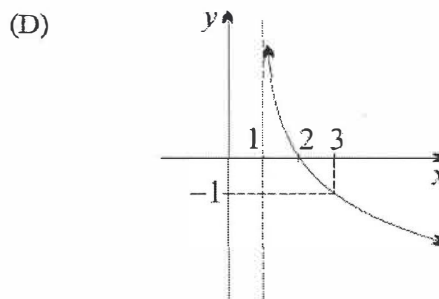
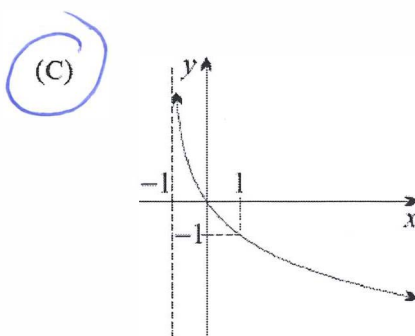
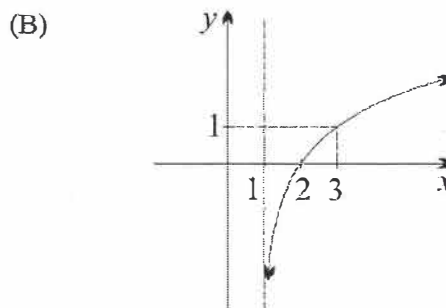
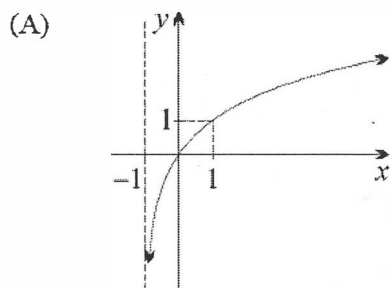
(D) $\frac{2\sqrt{2} + 1}{\sqrt{2}}$

$$\begin{aligned} \cos 135^\circ + \operatorname{cosec} 60^\circ &= -\cos 45^\circ + \frac{1}{\sin 60^\circ} \\ &= -\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{3}} \\ &= \frac{-\sqrt{3} + 2\sqrt{2}}{\sqrt{6}} \end{aligned}$$

- (b) Circle your correct alternative.

1

Which of the following is the graph of $y = -\log_2(x+1)$?



Question 2 Continued

- (c) Solve the equation $|2x - 3| = 6$.

2

$$2x - 3 = 6$$

$$\text{or } 2x - 3 = -6$$

$$2x = 9$$

$$2x = -3$$

$$x = 9/2$$

$$\text{or } x = -3/2$$



- (d) Differentiate with respect to x . Leave your answer in fully factorised form.

2

$$y = \frac{2x^2}{2-x}$$

$$u = 2x^2$$

$$v = 2-x$$

$$u' = 4x$$

$$v' = -1$$

$$y' = \frac{v u' - u v'}{v^2}$$

$$= \frac{(2-x) \times 4x - 2x^2 (-1)}{(2-x)^2}$$



$$= \frac{8x - 4x^2 + 2x^2}{(2-x)^2}$$

$$= \frac{8x - 2x^2}{(2-x)^2}$$

$$= \frac{2x(4-x)}{(2-x)^2}$$



End of Question 2

Question 3 (4 marks)

Marks

- (a) Solve $\sin\left(\frac{x}{2}\right) = \frac{1}{2}$ for $0^\circ \leq x \leq 360^\circ$

2

$$\sin\left(\frac{x}{2}\right) = \frac{1}{2}$$



$$\text{ref } \angle = 30^\circ$$

$$0 \leq x \leq 360^\circ$$

$$\frac{x}{2} = 30^\circ, 150^\circ \text{ (1)}$$

$$0 \leq \frac{x}{2} \leq 180^\circ$$

$$x = 60^\circ, 300^\circ \checkmark \checkmark$$

- (b) Show that $\cot A + \frac{\sin A}{1 + \cos A} = \operatorname{cosec} A$.

2

$$\text{LHS} = \cot A + \frac{\sin A}{1 + \cos A}$$

or

$$= \frac{\cos A}{\sin A} + \frac{\sin A}{1 + \cos A}$$

$$= \frac{\cos A}{\sin A} + \frac{\sin A \times (1 - \cos A)}{1 + \cos A \times (1 - \cos A)}$$

$$= \frac{\cos A(1 + \cos A) + \sin^2 A}{\sin A(1 + \cos A)} \text{ (1)}$$

$$= \frac{\cos A}{\sin A} + \frac{\sin A(1 - \cos A)}{1 - \cos^2 A} \text{ (1)}$$

$$= \frac{\cos A + \cos^2 A + \sin^2 A}{\sin A(1 + \cos A)}$$

$$= \frac{\cos A + \sin A(1 - \cos A)}{\sin^2 A}$$

$$= \frac{\cancel{1} + \cos A}{\sin A(\cancel{1} + \cos A)}$$

$$= \frac{\cos A + 1 - \cos A}{\sin A}$$

$$= \operatorname{cosec} A$$

$$= \frac{1}{\sin A}$$

$$= \text{RHS}$$

$$= \operatorname{cosec} A \checkmark$$

$$= \text{RHS}$$

End of Question 3

Question 4 (4 marks)

Marks

- (a) i) Show that the derivative of $f(x) = x^2 + 3x + 5$ is $f'(x) = 2x + 3$ by first principles.

2

$$f(x) = x^2 + 3x + 5$$

$$f(x+h) = (x+h)^2 + 3(x+h) + 5$$

$$= x^2 + 2xh + h^2 + 3x + 3h + 5$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

must have on every line

$$= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2 + 3x + 3h + 5) - (x^2 + 3x + 5)}{h} \quad (1)$$

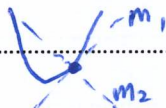
$$= \lim_{h \rightarrow 0} \frac{h(2x + h + 3)}{h}$$

$$= 2x + 3 \text{ as required}$$

must show 'h' cancels for 2 marks

- ii) Hence, find the equation of the normal to the curve $f(x) = x^2 + 3x + 5$ at the point $(1, 9)$.

2



$$f'(x) = (2x + 3)$$

$$f(1) = 5$$

$$m_1 m_2 = -1$$

$$m_2 = -\frac{1}{5} \quad (1)$$

$$y - y_1 = m(x - x_1)$$

$$y - 9 = -\frac{1}{5}(x - 1)$$

$$y - 9 = -\frac{1}{5}x + \frac{1}{5}$$

$$y = -\frac{1}{5}x + 9 + \frac{1}{5}$$

$$y = -\frac{1}{5}x + \frac{46}{5}$$

$$5y = -x + 46$$

$$5y + x - 46 = 0$$

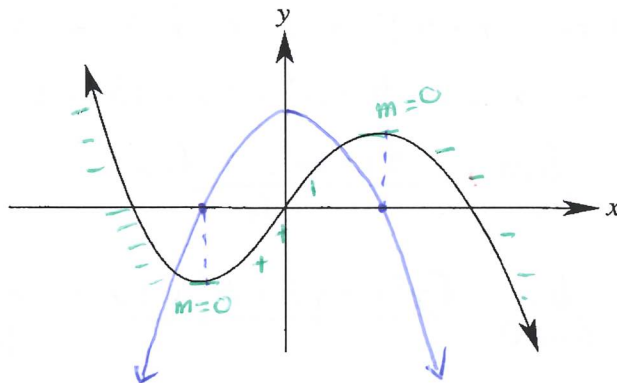
End of Question 4

Question 5 (4 marks)

Marks

- (a) The diagram below shows a sketch of the function $y = f(x)$.
On the same diagram sketch the gradient function $y = f'(x)$.

1



- (b) Show that, $\frac{d}{dx} [3x^2(4x-1)^3] = 6x(10x-1)(4x-1)^2$.

3

$$y = 3x^2(4x-1)^3$$

$$u = 3x^2 \quad v = (4x-1)^3$$

$$u' = 6x \quad v' = 3(4x-1)^2 \times 4 = 12(4x-1)^2$$

①

$$\frac{d}{dx} [3x^2(4x-1)^3] = uv' + vu'$$

$$= 3x^2 \times 12(4x-1)^2 + (4x-1)^3 \times 6x \rightarrow \text{①}$$

$$\Rightarrow 6x(4x-1)^2(6x + 4x-1)$$

$$= 6x(4x-1)^2(10x-1)$$

✓✓✓

factorising should be there for 3rd mark.

End of Question 5

Question 6 (4 marks)

Marks

- (a) Differentiate with respect to x .

$$y = \sqrt{x^2 - 2x}$$

2

$$y = \sqrt{x^2 - 2x}$$

$$y = (x^2 - 2x)^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2} (x^2 - 2x)^{-1/2} (2x - 2) \quad (1)$$

$$= \frac{1}{2 \sqrt{x^2 - 2x}} \cdot 2(x - 1)$$

$$= \frac{x - 1}{\sqrt{x^2 - 2x}}$$

✓✓

or

$$(x - 1) \cdot (x^2 - 2x)^{-1/2}$$

- (b) Solve $(\log_2 x)^2 - 8 = \log_2 x^2$.

2

$$(\log_2 x)^2 - 8 = \log_2 x^2$$

$$(\log_2 x)^2 - 8 - 2 \log_2 x = 0$$

$$u = \log_2 x$$

$$u^2 - 2u - 8 = 0$$

$$(u - 4)(u + 2) = 0$$

(1)

$$\therefore \log_2 x = 4 \quad \text{or} \quad \log_2 x = -2$$

$$x = 2^4$$

as

$$x = 2^{-2}$$

$$x = 2^4$$

or

$$x = \frac{1}{4}$$

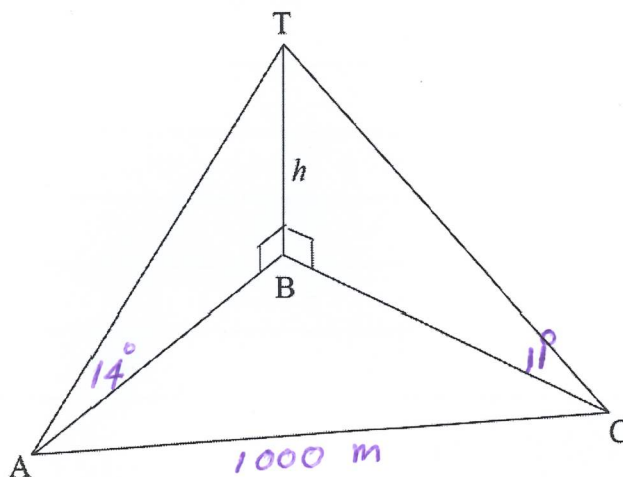
✓✓

End of Question 6

Question 7 (4 marks)

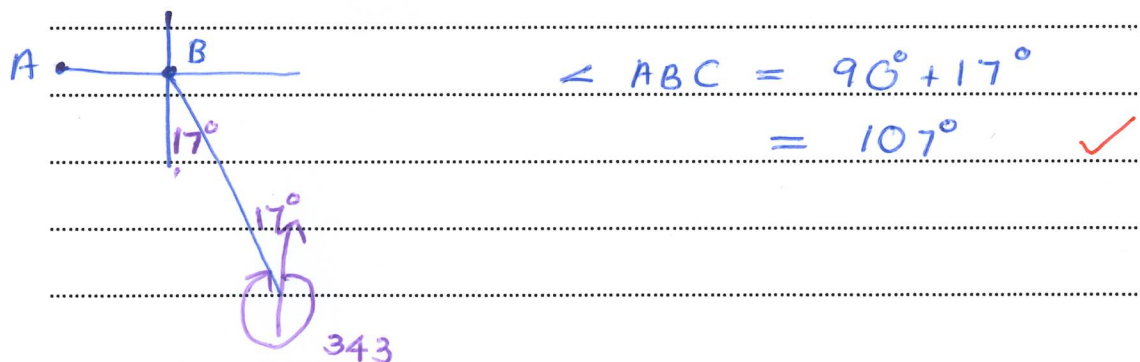
Marks

The angle of elevation of a vertical tower BT of height h metres, at a point A due west of it, is 14° . From the point C, the bearing of the tower is 343° and the angle of elevation is 11° . The points A and C are 1000m apart and on the same level as the base B of the tower.



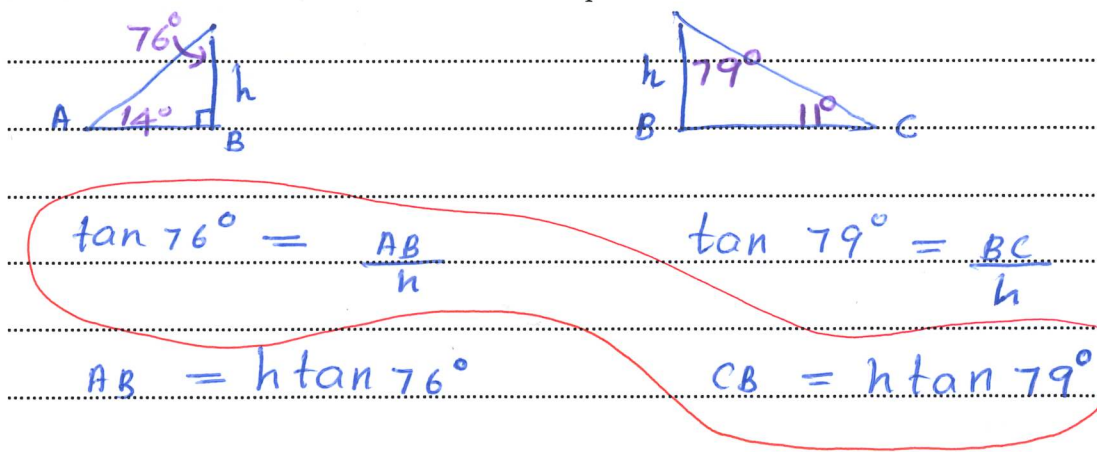
- (a) By drawing a diagram showing the bearings, explain why $\angle ABC = 107^\circ$.

1



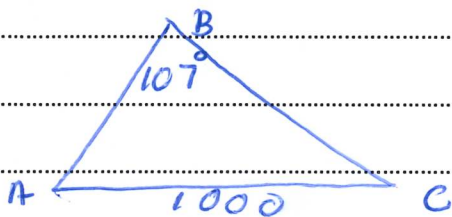
- (b) Show that $AB = h \tan 76^\circ$ and find a similar expression for CB .

1



- (c) Hence, find the height of the tower, correct to the nearest metre.

2



$$\begin{aligned} 1000^2 &= AB^2 + BC^2 - 2 AB \times BC \times \cos 107^\circ \\ &= h^2 \tan^2 76^\circ + h^2 \tan^2 79^\circ - 2 h^2 \tan 76^\circ \tan 79^\circ \cos 107^\circ \\ &= h^2 (\tan^2 76^\circ + \tan^2 79^\circ - 2 \tan 76^\circ \tan 79^\circ \cos 107^\circ) \end{aligned}$$

$$h^2 = \frac{1000^2}{\tan^2 76^\circ + \tan^2 79^\circ - 2 \times \tan 76^\circ \tan 79^\circ \cos 107^\circ}$$

$$h = 135.310$$

$$h \approx 135 \text{ m} \quad \checkmark \checkmark$$

End of Question 7

End of Assessment Task

